

SHAMILA FARHA S V

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Python Developer with strong foundational skills in building large-scale data processing pipelines, software automation frameworks, and robust database architectures. Demonstrates advanced proficiency in the Python software stack (NumPy, Pandas, Scikit-learn, PyTorch) alongside hands-on experience driving full-cycle development, structural code reviews, and comprehensive unit testing. Motivated postgraduate candidate with an analytical problem-solving mindset and a strong willingness to learn, adapt, and scale enterprise applications.

Technical Skills:

- **Programming & Scripting:** Python, Java
- **AI/ML & Data Engineering Frameworks:** NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Keras
- **Databases & Storage Layers:** SQL (MySQL), NoSQL (MongoDB), Relational Query Optimization
- **Web & APIs:** REST APIs, Telegram Bot API, Frontend Fundamentals (HTML/CSS/JavaScript wrappers)
- **Tools & Developer Software:** Visual Studio Code, Docker, Git, Streamlit, Raspberry Pi
- **Testing & Quality Engineering:** Code Review, Debugging, Troubleshooting, Unit Testing, Integration Testing, Test-Driven Development (TDD)

Soft Skills:

- Communication and Teamwork
- Analytical Problem Solving
- Time Management
- Adaptability & Continuous Learning

Professional Experience

SafeGuard Automated Data Pipeline & Software Framework

Role: Project Coordinator

Responsibilities:

- Designed, reviewed, and deployed a fully autonomous data processing framework in Python, integrating multi-threaded daemon queues to handle concurrent data transactions under a 100ms real-time latency threshold.
- Created, documented, and connected robust REST-style communication layers via the Telegram Bot API to dispatch live application tracking packets safely.

- Managed NoSQL data storage layers using MongoDB, optimizing query paths and schemas to ensure large-scale log persistence and real-time retrieval.
- Conducted regular code debugging and system troubleshooting, implementing structural test workflows to validate pipeline integrity and resolve runtime errors.

Impact:

- Improved system communication transparency and execution speeds by building clean, modular, and reusable API layers.
- Successfully maintained application process scheduling, preventing edge computing crashes and ensuring safe parallel processing workloads.
- Deepened practical knowledge of end-to-end Python software cycles, structural error isolation, and NoSQL database management.

2024 - 2025

College Project: Personal Helper Architecture & Predictive ML Pipeline

Role: Project Coordinator

Responsibilities:

- Built end-to-end machine learning and data processing pipelines on clinical data subsets using Scikit-learn, Pandas, and NumPy, achieving an 88% overall validation accuracy.
- Designed and structured relational database tables, optimized complex SQL transactional queries, and managed user profile metrics utilizing MySQL.
- Wrote comprehensive functional test cases, unit tests, and validation scripts to evaluate predictive model stability and rule out overfitting.
- Identified, isolated, and fixed interactive application display conflicts and backend bugs during integration testing.

Impact:

- Replaced manual, error-prone testing workflows by setting up automated model validation loops.
- Enhanced core competency in data preparation steps, relational database tuning, and hyperparameter engineering.
- Delivered all cross-functional project modules successfully within strict academic milestones.

2023 - 2024

Awards & Accomplishments:

- Oracle Certified Generative AI Professional & OCI AI Foundation Associate (2025)
- Introduction to Responsible AI Graduate Certification – Google Cloud (2024)
- National Service Scheme (NSS): Active Volunteer Recognition for structured civic service (2021 - 2024)
- Scouts & Guides Discipline and Leadership Badge (2019 - 2021)

Education

Master of Science in Computer Science (AI & ML) <i>Rajiv Gandhi National Institute of Youth Development (RGNIYD)</i>	2024 - 2026 <i>(Expected)</i>
Bachelor of Science in Computer Science <i>Peekay CICS Arts and Science College CGPA: 6.275/10</i>	2021 - 2024

Languages

English - Professional Proficiency
Malayalam - Native Proficiency

Interests

- Data Engineering & Pipelines
- Exploring Machine Learning Frameworks
- Scalable Application Architecture